

Human-AI Collaboration in Music Streaming and Production

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Abstract

This research paper examines two AI models—collaborative filtering (CF) and generative adversarial networks (GANs)—used in human-AI collaboration within music streaming services and production tools. It explains how these algorithms operate and are applied, particularly through unsupervised learning, and evaluates their performance metrics. It considers the strengths and limitations of these algorithms. CF and GANs demonstrate the potential of AI to enhance human creativity by facilitating music search, enhancing the listening experience, and supporting composition.

Glossary of terms

- **HAIC – Human-AI Collaboration**
- **ML – Machine Learning**
- **GAN – Generative Adversarial Network**
- **CF – Collaborative Filtering**
- **ALS – Alternating Least Squares**
- **DAW – Digital Audio Workstation**
- **GAW – Generative Audio Workstation**
- **NDCG - Normalized Discounted Cumulative Gain**

Introduction

Human-AI Collaboration (HAIC) is a developing field of AI that examines the methods by which humans and artificial intelligence collaborate to achieve results that exceed their complementarities (*Human-AI Collaboration – IBME*, n.d.). It is a recent advancement that has allowed individuals to utilise AI tools, especially Generative AI (GenAI), for the creation of diverse content types, including text, audio, and visual materials(Pophal, n.d.). With the aid of Machine Learning (ML) and Natural Language Processing models, these systems collect and train on large amounts of data to support human creativity. Streaming platforms, such as Spotify’s recommendation engine and YouTube Music’s personalised mixes, which are tailored to users’ behaviour, taste and moods(Mokoena & Obagbuwa, 2024)(see Figure 1).

In production, Digital Audio Workstations (DAWS) are evolving into Generative Audio Workstations (GAWs) – systems that not only record and organise sound but also assist in generating, enhancing, and proposing musical ideas. (From DAW to GAW, 2025). Research on HAIC in music not only reveals technological innovations but also challenges conventional notions of authorship, artistic creativity, and originality. As artificial intelligence system assumes responsibilities in composition and decision-making, the artist’s role changes from that of a solo creator to a co-creator or collaborative partner, raising questions about creativity, originality, and artistic identity in an AI-driven era(Maravilla et al., 2024).

The remainder of the report will discuss the chosen AI algorithms, specifically Collaborative Filtering and Generative Adversarial Networks, which are the two primary algorithms used in music streaming and production. Explaining how the algorithms work and their applications in the real world. Next, the Results and Analysis section will assess the performance of these algorithms in real-world applications, utilising criteria such as accuracy and user engagement—the strengths and weaknesses in practical applications, and broader implications encompassing ethical considerations. The conclusion summarises the lessons and explores future potential for human-AI co-creativity in music.

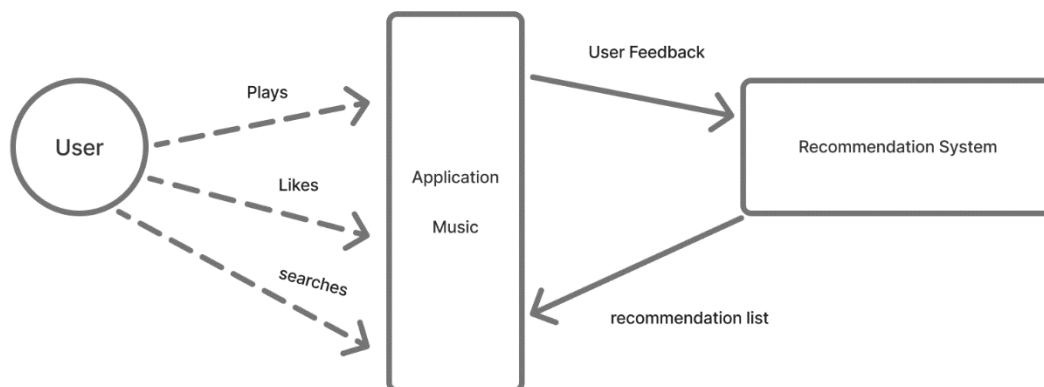


Figure 1: Example of an AI-based recommendation process in music streaming platforms(Yubo, 2021)

Chosen AI Algorithm

Application of Collaborative Filtering in Spotify

In this section, CF is applied in Spotify for personalised music recommendation and playlist generation. Machine learning (ML) models can only be effective as the data on which its been trained. Spotify's comprehensive data process is a major factor that contributes to its accurate recommendation algorithms. Two main types of data are being gathered: artist-provided metadata and raw audio data (Wong, 2024). Metadata includes details such as song title, artist name, album, and genre, while raw audio data captures technical characteristics like beats per minute (BPM), loudness, and musical keys. As this data is also collected, Spotify uses natural language processing (NLP) models to analyse lyrics, user-generated playlist names, and listener comments across the internet to infer contextual meaning(Wong, 2024).

Collaborative Filtering (CF) uses the K-nearest neighbour (KNN) technique, which helps calculate the similarities between users based on listening history. This means CF learns users' tastes from user-item interactions, like a friend sending music with similar preferences (Madathil, 2017). CF methods are divided into three categories: memory-based, model-based, and hybrid. Memory-based filtering predicts items from past ratings or interactions, employing algorithms such as item-based top-N recommendations to find users with common or shared tastes in music(Madathil, 2017). Model-based filtering uses ML to detect user preference patterns and make predictive recommendations, with limited data provided. While the hybrid is the combination of memory– and model-based methods to increase the accuracy and address the method's limitations.

This system operates in a continuous loop, where it can identify users with similar interests (neighbours) and recommend music based on their tastes(Madathil, 2017). However, the user receives those recommended songs and provides feedback through actions such as searches, likes or plays. The endless cycle allows the system to understand from the user evaluations, enhance neighbour identification, and continuously improve the results of future recommendations (Madathil, 2017). See Figure 2 for the collaborative filtering flowchart.

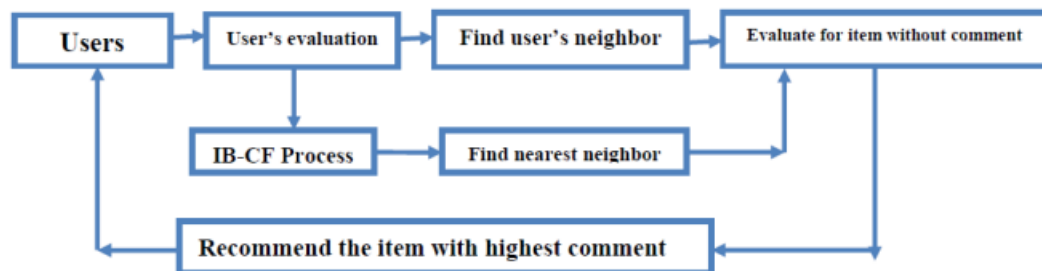


Figure 2: Collaborative Filtering Recommendation System(Madathil, 2017)

Another way to represent this process is by using a user-item matrix. In the matrix, each row represents a unique user, and each column represents a song or an artist. At the intersection of each row and column pairing lies a score that shows details of how much the user (row value) likes a certain song/artist (column value). The score can comprise various variables like repeated listens, playlist saves, skips, and total listening duration(Wong, 2024).



Figure 2.1: Collaborative filtering process illustrated through a user-item matrix representation (Wong, 2024)

The technical process of using the user-item matrix is shown in Figures 2.2, 2.3 & 2.4. The model posits that a user's song preference can be expressed as the product of two latent vectors: a user vector ($\mathbf{X}_u$) and an item/song vector ($\mathbf{Y}_i$). The system uses implicit feedback, like whether a user has listened to a song, to train these vectors instead of explicit ratings (Madathil, 2017). The binary variable $\mathbf{P}_{ui}$ tells you if user u has done something with item i (e.g. streamed a song). $\mathbf{P}_{ui} = 1$ if the user has heard the song at least once; otherwise, $\mathbf{P}_{ui} = 0$. Then, based on how much the user interacted with that item, a confidence score $\mathbf{C}_{ui}$ is given. The goal is to find the best user vector ($\mathbf{X}_u$) and item vector ($\mathbf{Y}_i$) so that their dot product is close to what the user wants. To stop overfitting, a regularisation term is added.

$$\min_{x_u, y_i} \sum_{u,i} c_{ui} (p_{ui} - x_u^T y_i)^2 + \lambda \left(\sum_u \|x_u\|^2 + \sum_i \|y_i\|^2 \right)$$

Figure 2.2: Minimise cost function (Madathil, 2017)

$$x_u = (Y^T C^u Y + \lambda I)^{-1} Y^T C^u p(u)$$

Figure 2.3: Item Vector (Madathil, 2017)

$$y_i = (X^T C^i X + \lambda I)^{-1} X^T C^i p(i)$$

Figure 2.4: User Vector (Madathil, 2017)

To solve this optimisation problem, the system uses Alternating Least Squares (ALS). First, all item vectors (Figure 2.2) are fixed while solving for optimal user vectors, and the item vectors are updated. This alternating process continues until the model converges and the cost function can no longer be reduced.

This algorithm primarily operates through unsupervised learning, as the models are not provided with explicit labels indicating what a user would like. Instead, user interactions are interpreted as implicit positive signals of preference, while songs that are never interacted with are treated as unknown rather than negatives (Mccallum et al., 2022.). It learns hidden patterns by attempting to identify these underlying interactions. It usually optimises either a reconstruction goal (reconstructing the known user-item interaction matrix from learnt latent factors) or a ranking goal. To address the class imbalance and prevent the model from recommending music that everyone likes, techniques such as negative sampling or weighting are employed, as only a few songs receive explicit positive interactions from users out of thousands of possible tracks.

Application of Generative Adversarial Networks (GANs) in Music Production

A generative adversarial network (GAN) is a machine learning model that learns patterns from existing training data and generates new, realistic data based on those patterns (Varughese, n.d.). It is a deep learning model used in music production and creative assistance. GAN consists of two neural networks: the generator, which creates new data like audio waveforms or MIDI sequences, and the discriminator, which assesses the authenticity of the output as either real or artificially created (Feng Liu, 2024) (Charles et al., 2019).

Through adversarial training, the generator enhances its capabilities by deceiving the discriminator, allowing the system to generate authentic musical outputs. GANs have been utilised in both audio and symbolic: WaveGAN and GANSynth produce raw audio, such as drum hits and instrument tones, and MuseGAN creates multi-instrument compositions, see figure 3 showing a proposed MuseGAN model (Hao-Wen Dong et al., 2024). GANs are good for music production because of their model complexity, high-dimensional data distributions without labelled datasets, making the training process unsupervised.

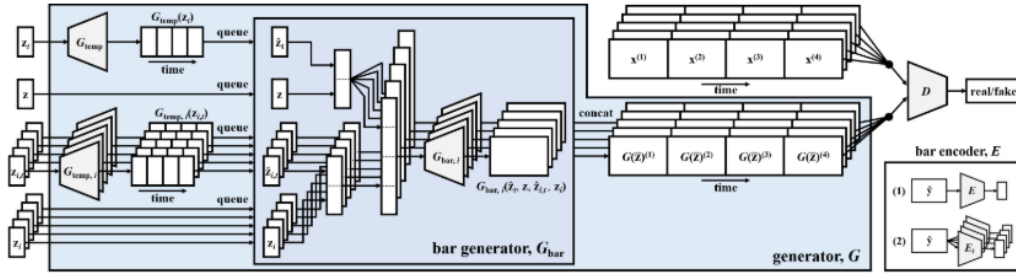


Figure 3: Diagram of the proposed MuseGAN model for multi-track sequential data generation (Hao-Wen Dong et al., 2024)

MuseGAN builds on the multi-track and temporal models by combining them into an architecture that generates multi-instrument music (Hao-Wen Dong et al., 2024). In Figure 3, the input to MuseGAN, represented by $\mathbf{z}$, is made up of four parts: an inter-track time-independent random vector $\mathbf{z}$, an intra-track time-independent random vector $\mathbf{z}_i$, an inter-track time-dependent random vector $\mathbf{z}_t$ and an intra-track time-dependent random vector $\mathbf{z}_{i,t}$ (Hao-Wen Dong et al., 2024). Each instrument track uses two temporal generators: one shared across all tracks and one specific to the track. It provides time-varying random vectors and produces latent representations that capture overall musical timing and track-specific rhythmic patterns. These latent vectors are then combined with time-independent random inputs and fed into a bar generator, which sequentially produces the final piano-roll music output bar by bar (Hao-Wen Dong et al., 2024). Figure 3.1 will show the formula for this process (Hao-Wen Dong et al., 2024).

$$G(\mathbf{z}) = \left\{ G_{\text{bar}, i} \left(\mathbf{z}, G_{\text{temp}}(\mathbf{z}_t)^{(t)}, \mathbf{z}_i, G_{\text{temp}, i}(\mathbf{z}_{i,t})^{(t)} \right) \right\}_{i,t=1}^{M,T}.$$

Figure 3.1: MuseGAN Generation Function (Bar-Level Multi-Track Generator Formula) (Hao-Wen Dong et al., 2024)

In human-AI collaboration, GANs serve as creative partners, producing melodies, tunes or musical patterns that artists or producers can refine. Despite their potential, GANs have a significant amount of training challenges and encounter problems such as mode collapse or restricted controllability.

Result and Analysis

Collaborative Filtering (CF) in music streaming is assessed using accuracy metrics such as NDCG, Precision@N and Recall@N, with user engagement indicators including listening duration, skip rates, and retention (Vall et al., 2017). According to Vall et al. (2017), a basic CF model achieves about 35% Recall@100 for songs that show frequently on training playlists. However, for rare songs (occurring in ≤ 4 playlists), the CF recall@100 significantly decreased to approximately 3%. See Table 1 to show more results on this analysis. Figure 4 illustrates how Spotify's Discover Weekly playlist is generated. Spotify's Discover Weekly showcases exceptional performance, enhancing both listening hours and user pleasure. Diversity and catalogue comprehensiveness are crucial for sustained engagement (Madathil, 2017).

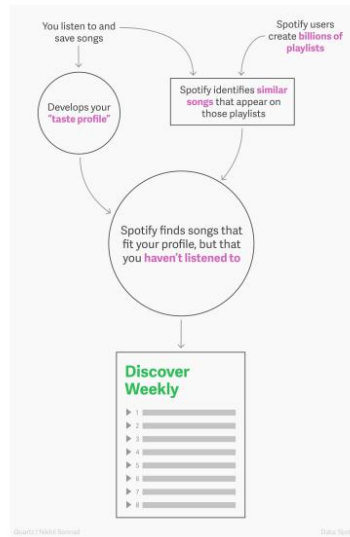


Figure 4: Illustrates Spotify's Discover Weekly Playlist Flow Chart (Wong, 2024)

Performance Indicator	CF (Music Streaming) Key Finding
Recommendation Accuracy (Recall@100)	~35% for frequent songs, dropping to ~3% for rare songs with pure CF (Vall et al., 2017). The Hybrid models increase the accuracy of the recall on long-tail terms (Vall et al., 2017).
User Engagement (Spotify Discover Weekly)	>2.3 billion hours streamed over the past 5 years; Discover Weekly users stream 2x longer than others (Spotify Advertising Team, 2020), leading to a powerful engagement

Table 1: Summarises some key performance findings for CF-based recommenders

GAN models such as MuseGAN produce multi-track musical concepts that listeners evaluate as musically coherent, achieving an average score of approximately 3.3/5 – superior to simpler GAN baselines, although still falling short of human-level organisation (Hao-Wen Dong et al., 2024). Table 2 shows the results of this study. GANSynth produces high-fidelity audio and is favoured over WaveNet-based systems in blind hearing evaluations, while also being over 50,000 times faster in generation, rendering it appropriate for real-time creative processes (Engel, 2019). Blind listening test confirmed that listeners prefer the GANSynth audio, especially the IF-Mel + HP configuration, delivering higher fidelity and a more natural timbre (Engel, 2019) (See figure 5 for a visual representation of listener preference across models.) GANs face challenges with long-term coherence, mode collapse and stylistic bias, necessitating that human artists frequently curate and adjust the outputs rather than utilising them directly (Hao-Wen Dong et al., 2024).

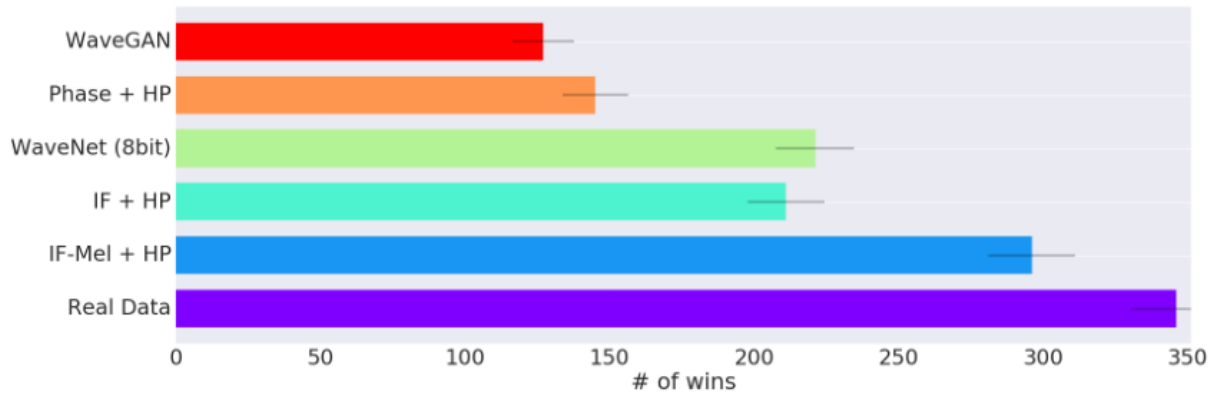


Figure 5: Listener preference comparison between GANSynth and baseline audio models (WaveGAN and WaveNet Autoencoder)(Engel, 2019)

			H	R	MS	C	OR
from scratch	non-pro	jam.	2.83	3.29	2.88	2.84	2.88
		comp.	3.12	3.36	2.95	3.13	3.12
		hybrid	3.15	3.33	3.09	3.30	3.16
	pro	jam.	2.31	3.05	2.48	2.49	2.42
		comp.	2.66	3.13	2.68	2.63	2.73
		hybrid	2.92	3.25	2.81	3.00	2.93
track-conditional	non-pro	jam.	2.89	3.44	2.97	3.01	3.06
		comp.	2.70	3.29	2.98	2.97	2.86
		hybrid	2.78	3.34	2.93	2.98	3.01
	pro	jam.	2.44	3.32	2.67	2.72	2.69
		comp.	2.35	3.21	2.59	2.67	2.62
		hybrid	2.49	3.29	2.71	2.73	2.70

Table 2: User study results (**H**: harmonious, **R**: Rhythmic, **MS**: musically structured, **C**: coherent, **OR**: overall rating)(Hao-Wen Dong et al., 2024)

Conclusion

This paper examined how HAIC can be applied in music, using two distinct models: Collaborative Filtering (CF) in music streaming and Generative Adversarial Networks (GANs) in production. CF increases accuracy in music discovery and user engagement by predicting preferences through its risk of over personalization. GANs give room for creative music production and give broader access as well, while facing challenges like consistency and bias. Both models operate on unsupervised learning and are evaluated using relevant metrics. While each approach has strength and limitations, these algorithms represent how AI can augment creativity. This duality shows the evolving role of AI as both a curator and creator in music streaming and production.

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